

NPLify · P0 Technical Baseline

Deliverable 4 — Transaction Lifecycle & Approval State Machines · Draft v1.0

New XP Technologies Limited

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Status: Built against ERD & Data Model Draft v4.0, the Ledger Posting Design v1.0 and the Project Understanding v1.0 with NPL's review comments. It defines the states a transaction passes through from quote to settlement, the events that move it between states, and for every controlled action who initiates it, who approves it, when it escalates to Management and what happens on rejection.

Audience: NPL Operations, Finance and Management, and the New XP engineering team. Figures show the normal path in blue, exceptions in orange, rejections dashed in red, and end states as rounded boxes.

1 · Three roles, one principle

Every controlled action is **maker-checker**: one person initiates, a different person in the approving role approves, and the system refuses an approval by the initiator. Visibility is cumulative: Operations sees the least, Finance sees everything Operations sees plus the economics, Management sees everything.

Role	Does	Sees	Never sees
Operations	Quotes, verifies collections, instructs conversions, prepares disbursements, requests reroutes, records partner approvals	Operational fields, sender-facing prices, partner rates as entered, deal states	Reference rate at capture, spread, margin, pool, cost breakdown, ledger balances, variances, earnings — on any screen, export, message or report
Finance	Approves what Operations initiates (quotes, rates, conversions, settlements, exceptions); resolves variance exceptions; owns reconciliation and the Finance Log; proposes configuration	Everything Operations sees plus all economics, postings and balances	—
Management	Approves escalations and the heavier actions listed in Section 4; approves configuration and user changes; countersigns releases when only one Finance user is on shift	Everything	—

Where the Understanding document says two things. Section 7.1 says a reroute is "Ops requests, Finance approves"; Section 9 lists "contingency routes" among Management's approvals. Both are kept: Finance approves the individual reroute; Management approves the *set* of alternate partners a project may

reroute to (PARTNER_CONFIG.applicable_for_reroute, which is configuration) and, together with Finance, every outbound transfer from NPL's own wallet.

2 · The deal: Quote → Collect → Convert → Settle

A deal is a group of one or more legs. A quote-first leg walks the whole top row of Figure 1; a collect-first leg is born at Collected because the money arrived before any quote; an estimate-first leg enters Collecting with an indicative figure and is priced at conversion; a return leg is born at Collected, funded by the returned payout. Pricing fields are stamped once and never recomputed.

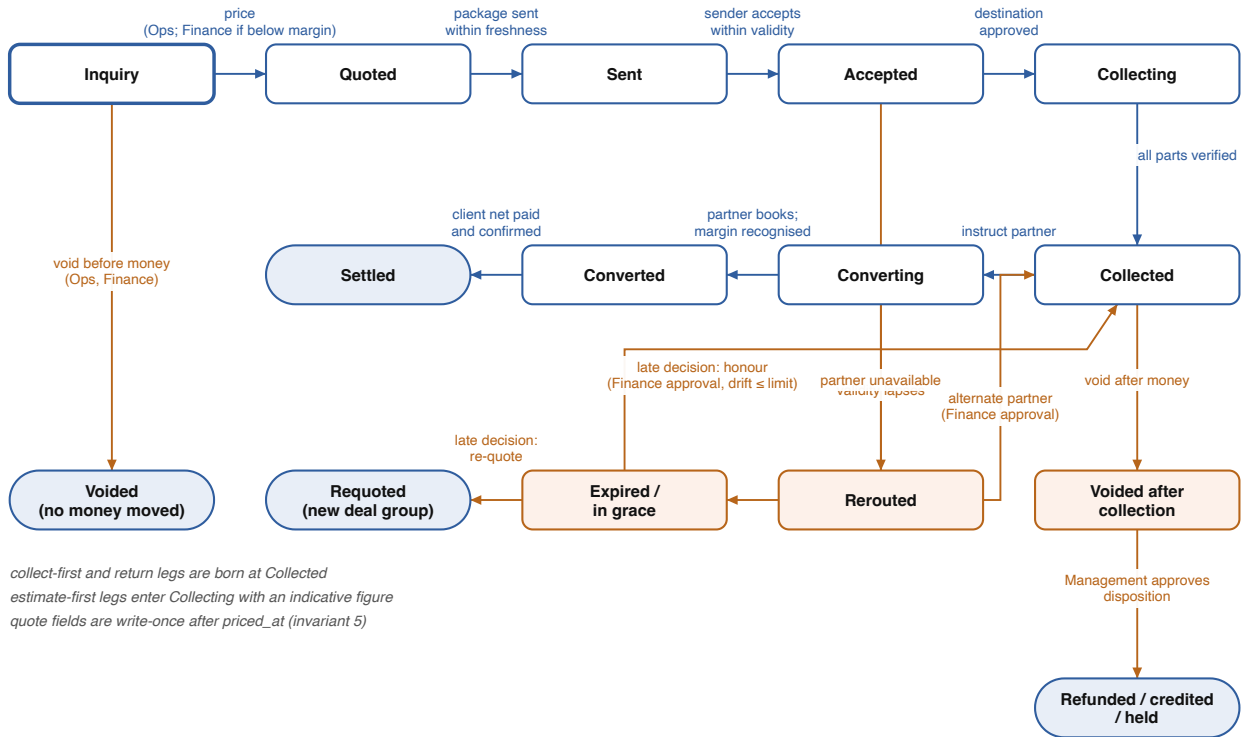


Figure 1 — Deal leg lifecycle. Blue: normal path. Orange: exceptions. Rounded: end states. Collect-first and return legs enter at Collected; estimate-first enters at Collecting.

Transitions

From	To	Event	Guard	Who
Inquiry	Quoted	Price computed: partner × pair chosen, fee structure applied, rate snapshot taken, day's rate comparison created if missing	Partner rate version usable (not breached); margin ≥ floor, else Finance approval; an approved destination registration exists for the entitled group with that partner (invariant 8)	Ops; Finance if below margin
Quoted	Sent	Quote package marked sent	Within quote_freshness_min of the snapshot; otherwise re-price	Ops
Sent	Accepted	Sender accepts	Within quote_validity_min	Ops records
Sent / Accepted	Expired / in grace	Validity lapses	Grace applies for grace_min only if market drift ≤ grace_drift_pct	System
Expired / in grace	Requoted	Late decision: re-quote	New deal group reference; the original stamp is kept	Ops

From	To	Event	Guard	Who
Expired / in grace	Collected	Late decision: honour the old rate	Project late_collection_policy allows it; recorded Finance approval	Finance
Accepted	Collecting	Instructions issued (wallet, bank-in details or cash instructions)	—	Ops
Collecting	Collected	All parts verified (hash / slip / receipt)	Collected amount matches; overpayment → sender credit; underpayment → wait or void	Ops verifies
Collected	Converting	Conversion instructed to the partner	Rate version still usable; receiver fee taken at conversion	Ops; Finance approves
Converting	Converted	Partner books; actual_out recorded; margin recognised; whole-unit truncation applied	Variance within tolerance, else an exception opens (the deal does not wait)	System; Finance on exception
Converted	Settled	Every euro of client net paid out and confirmed (whole units; dust carried)	Disbursement lifecycle, Figure 2	—
Inquiry ... Collecting	Voided	Void before money moved	Reason recorded; no postings	Ops; Finance
Collected / Converted	Voided after collection	Void after money moved	Disposition: refund (network-fee policy), sender credit, or hold	Ops; Management
Converting / Collected	Rerouted → Collected at the alternate partner	Partner unavailable, limit reached, holiday, compliance hold	Approved REROUTE; alternate partner in the applicable set	Ops; Finance

3 · Settlement, rates, reroute and approval records

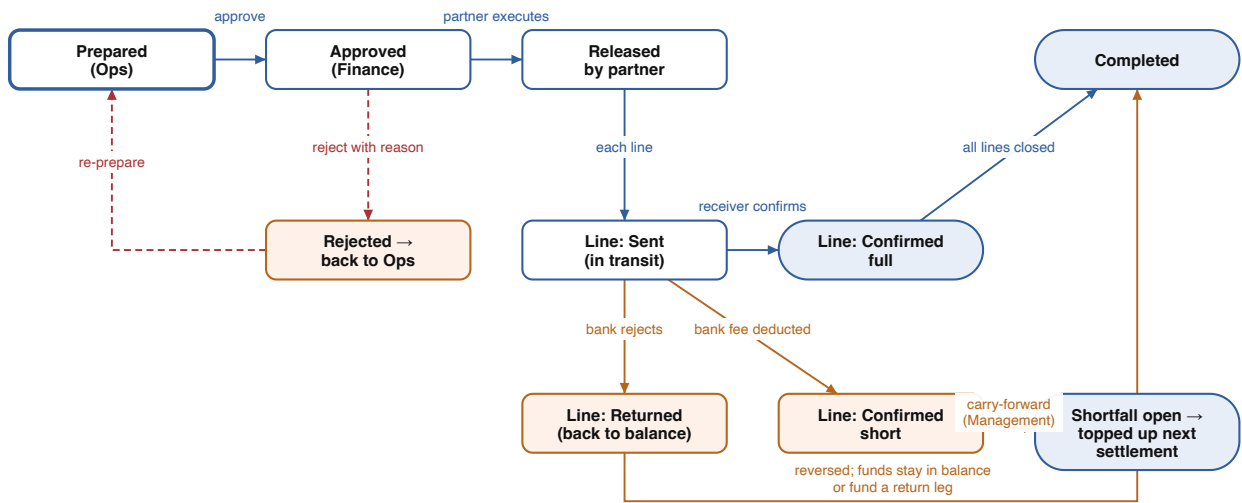


Figure 2 — Settlement: a disbursement and its lines. One disbursement completes when every line is confirmed or returned.

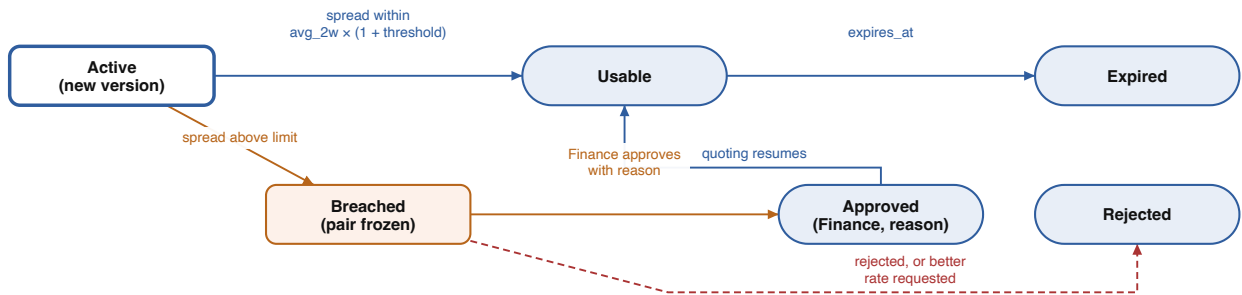


Figure 3 — Partner rate version. A breach freezes quoting and conversion on that partner × pair; there is no skip path.

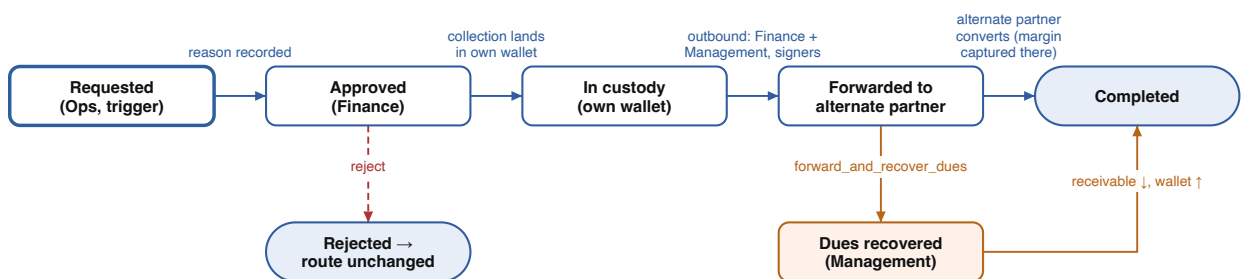


Figure 4 — Reroute. The only time NPL holds client money; every resting balance carries deal, reason and age.

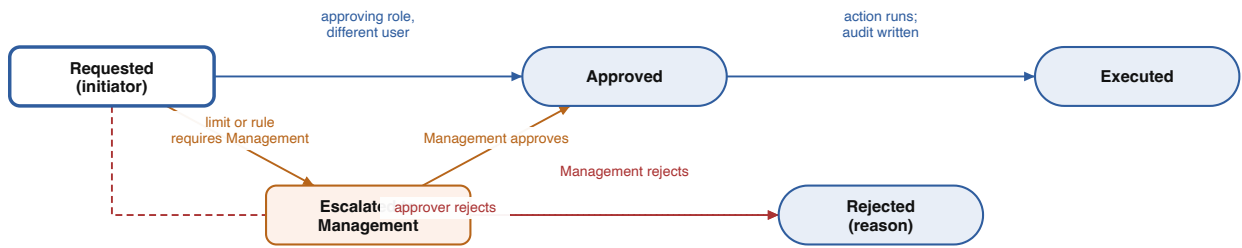


Figure 5 — An APPROVAL record. Approver $\neq$ initiator is enforced at the endpoint; a rejection never deletes anything.

4 · Controlled actions: who initiates, who approves, what rejection means

The approving role is always a different user from the initiator. "Escalates to Management when" lists the conditions that add a Management approval on top of, not instead of, the approving role's.

Controlled action	Initiates	Approves	Escalates to Management when	On rejection
Quote a deal (price, fee, partner × pair)	Operations	Finance	Below-margin quote: Finance approval mandatory before sending	Returns to Ops with reason; the quote package is not sent; Ops re-prices as a new quote
Honour an expired quote (late decision)	Operations	Finance	Project policy must allow honouring; market drift within grace_drift_pct	Re-quote at today's rate with a new deal reference; the original stamp is kept
Enter a manual rate (reference feed outage)	Operations	Finance	Evidence attached (both feeds down)	No rate; quoting waits for the feed or a new manual entry
Confirm a partner rate version	Operations (enters)	Finance (confirms)	Breached version: Finance approval with recorded reason, or rejection; no skip path	Pair stays frozen; Ops may request a better rate from the partner, which is a new version
Verify a collection (hash / slip / receipt)	Operations	— (audited; Finance reviews in reconciliation)	Underpayment left open beyond policy: Finance decides wait or void	Collection stays unverified; funds are not attributed
Instruct a conversion	Operations	Finance	Variance beyond tolerance opens an exception that Finance resolves	Conversion not instructed; deal stays Collected
Resolve a conversion variance exception	Finance	Finance (second user)	Variance above the project's exception limit → Management	Exception stays open; the deal is already converted and does not wait
Prepare and release a disbursement	Operations	Finance	Above settlement limit; third same-day slot; payout to a receiver in another group (offset); only one Finance user on shift → Management countersigns	Back to Prepared with reason; lines can be changed and re-submitted
Override the bank-fee treatment on one settlement	Operations	Finance	—	Project default applies
Top up a shortfall from the pool	Finance	Management	Always Management (company money becomes client money)	Shortfall stays open; receiver remains short
Pay a receiver in another receiver group (offset)	Operations	Finance	Always Management, with tighter aging on the resulting offset	Lines must stay within the entitled group
Request a reroute	Operations	Finance	Contingency route set (applicable_for_reroute) is configuration: Management	Deal stays on its route; if the partner is unavailable the deal waits or is voided

Controlled action	Initiates	Approves	Escalates to Management when	On rejection
Outbound transfer from NPL's own wallet	Finance	Management (dual approval)	Always; executed by NPL signers outside the platform (hardware or multisig)	Funds stay in custody; aging continues to be reported
Recover dues on a reroute (net against forwarded funds)	Finance	Management	Always; amount $\leq$ outstanding receivable at that partner; partner's agreement recorded	Forward the full amount
Balance conversion (client balance to another currency)	Finance	Management	Always; any excess over the served group's entitlement is an offset and needs the offset approval too	Balance stays in its currency
Open a return leg after a rejected payout	Operations	Finance	Destination other than the sender's own return receiver → Management with reason; markup apply / waive is a FEE_DECISION approved by Finance	Returned funds stay in balance for later payouts
Override the network-fee policy on a refund	Operations	Finance	—	Project default applies
Void a deal before money moved	Operations	Finance	—	Deal continues
Void a deal after collection (refund / sender credit / hold)	Operations	Management	Always	Deal continues; funds stay attributed
Record a partner's destination approval (SETTLEMENT_REGISTRATION)	Operations	Finance	—	Registration stays pending_partner; quote-first deals for it cannot leave Inquiry
Change thresholds, fee structures, partner configuration, rates sources	Finance	Management	Always; all versioned	Current version stays in force
Create, change or remove a user or role	Management	Management (second user)	Always	No change
Ledger adjustment (reversal and re-posting)	Finance	Management	Always; the reversal names the transaction it reverses	Books unchanged; the discrepancy stays on the exception list

5 · Rejection handling, the same way everywhere

1. **Nothing is deleted.** The request, the rejection and its reason stay on the APPROVAL record and in the audit log. A re-submission is a new approval request that points at the rejected one.
2. **The record returns to the initiator's last safe state.** A rejected quote is not sent; a rejected disbursement goes back to Prepared; a rejected reroute leaves the route unchanged; a rejected balance conversion leaves the balance in its currency.
3. **Money that already moved is never rolled back by a rejection.** A conversion the partner has booked stands; a variance exception is resolved by a decision and, if needed, a reversing ledger entry, not by editing the conversion.
4. **A rejection reason is mandatory** and is visible to the initiator; derived-economics fields are stripped from the reason shown to Operations.
5. **Frozen states need a decision, not time.** A breached rate version, an open variance exception, an open offset and custody in NPL's wallet all stay on the daily exception list until a role with authority acts.
6. **Escalation never skips a level.** Management approves in addition to Finance; when only one Finance user is on shift, Management's countersignature stands in for the second Finance user.

6 · State tables by entity

Entity	States and transitions	Notes
DEAL_GROUP	Inquiry → Quoted → Sent → Accepted → Collecting → Collected → Converting → Converted → Settled; Expired / in grace; Requoted; Voided; Voided after collection → Refunded / Credited / Held	The group's state follows its legs: Settled when every forward leg is settled and no return leg is open.
DEAL (leg)	As Figure 1. A collect-first or return leg is born at Collected; an estimate-first leg enters Collecting; a leg 2 of a route is born at Collected when leg 1 pays the transit receiver	leg_type and funding_source decide the entry point. Pricing fields are write-once after priced_at.
COLLECTION	Pending → Received → Verified; Rejected	Parts post as verified. Overpayment creates SENDER_CREDIT; underpayment waits or is voided.
CONVERSION	Expected → Booked → Approved; Variance exception (Booked, beyond tolerance) → Resolved	The booked figure is what the partner did; the exception is resolved by Finance; the deal does not wait.
BALANCE_CONVERSION	Requested (Finance) → Approved (Management) → Booked; Rejected	Owns exactly one CONVERSION; entitlement re-attributed to the served group on booking.
REROUTE	Requested → Approved → In custody → Forwarded → Completed; Rejected	Phase pre_collection or post_collection; outcome via_own_wallet or refund_sender.
DISBURSEMENT	Prepared → Approved → Released → Completed; Rejected (back to Prepared)	Completed when every line is confirmed or returned.
DISBURSEMENT_LINE (outcome)	sent → confirmed; sent → returned	A returned line is reversed into the balance before anything else happens.

Entity	States and transitions	Notes
CONFIRMATION (result)	full short	Short opens a SHORTFALL unless the treatment is waived.
SHORTFALL	open → topped_up	Top-up funded from the pool, Management approval, paid with the next disbursement.
BANK_FEE_EVENT	open → applied	Applied to the disbursement that carried the true-up.
DISBURSEMENT_RETURN	recorded (terminal)	Fund a return leg or stay in balance.
RATE_COMPARISON	open → closed	One per partner × pair × date; closed at end of day.
PARTNER_RATE_VERSION	active → usable → expired; active → breached → approved → usable, or rejected	Never overwritten; each conversion names its version.
SETTLEMENT_REGISTRATION (approval_status)	pending_partner → approved → suspended → approved	A quote-first deal leaves Inquiry only if the entitled group has an approved registration with the chosen partner.
FEE_DECISION	proposed → approved → applied; rejected	One subject (deal leg, disbursement or balance conversion); standing rules are unaffected.
SENDER_CREDIT	open → applied	Applied to a later deal or refunded (a void disposition).
EARNINGS_RECEIVABLE	outstanding → recovered	Recovered on the partner's fee cycle or by netting on a reroute.
REFERRAL_ACCRUAL	accrued → paid	Paid from NPL's earnings.
PARTNER_REBATE_ACCRUAL	expected → reconciled → received	Monthly against the partner's statement.
APPROVAL	requested → approved rejected; requested → escalated → approved rejected	As Figure 5.

7 · Aging and the daily exception list

Four things age and are reported daily with deal, reason and hours since the event: client money in NPL's own wallet (custody), open offsets across receiver groups, open shortfalls, and open variance exceptions. Breached rate versions and frozen partner × pairs appear on the same list. Time limits for each are project configuration (THRESHOLD) and are not fixed by this document.

Draft v1.0 — for review with NPL. State names are the proposal for the state columns the ERD leaves as free text; once agreed they become the enumerations of those columns.